The Riemann Hypothesis: Conclusio

Part V of a Five-Part Series

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Abstract

This paper concludes a five-part investigation of the Riemann Hypothesis through spectral analysis, arithmetic thermodynamics, and the Weil quadratic form:

Part I [5]: Foundations and Obstructions.

Part II [4]: Even Dominance (Branch A).

Part III [6]: Spectral Routes (Branches B and Z).

Part IV [3]: Cross-Route Synthesis.

Part V (this paper): Conclusio.

We present the condensed atlas: three independent conditional reductions of RH (Even Dominance, Hadamard Strip, CCM Microcluster), three structural identifications ($c_\Xi \approx 0.0231$, $C^* = 64\pi^2/3$ as a $\lambda = 5, 7$ boundary-moment benchmark with $\lambda = 11$ still open, $g_0(m) \approx 2\pi/\log \gamma_m$), two conditional theorems (B-1, GF1), and two isolated open proof slots. The series title remains "From Landscape to Atlas"; promotion to "From Atlas to Proof" is governed by a five-gate audit protocol triggered only by a substantive mathematical breakthrough.

Series status (v3.0, 2026-05-26; criticality-lens update 2026-05-31). Five-part series; all three conditional reductions unchanged from v2.2. Structural reorganisation only; no new proof claims. A 2026 working phase adds a *heuristic* criticality lens (RH $\Leftrightarrow \Lambda = 0$), a sharper common-wall localization, and a computable Laguerre–Pólya/Herglotz diagnostic (Part IV [3]); these re-weight priorities and close four *concrete constructions*, but make no new proof claim.

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1 Atlas Overview

Route	Sector	Status	Key result	Paper
A (Even Dom.)	Even (Part II)	conditional	A1–A8 chain + GF1 benchmark with $C^* = 64\pi^2/3$ for $\lambda = 5, 7$; $\lambda = 11$ open	[4]
B (Hadamard)	Spectral (Part III)	conditional	$A_\lambda \le 1.0063$ under log-curvature control	[6]
Z (CCM)	Spectral (Part III)	conditional	C2ca.1 + C2ca.2 + C2ca.5; $g_0(m) = 2\pi/\log\gamma_m$	[6, 7]
Cross-Route	Combined (Part IV)	empirical	shared Galerkin pattern; 2026 criticality lens + sharpened common wall ($\Re s = 1$ continuation)	[3]

2 Result Summary

2.1 Three Structural Identifications

1. **Hadamard curvature constant (RH-neutral):**

$$c_\Xi = -\frac{1}{2}\sum_\rho \frac{1}{(\rho - \frac{1}{2})^2} \approx 0.0231.$$

   Under RH, this specialises to $\sum_{k\ge1} 1/\gamma_k^2$. (Part III [6].)

2. **Boundary-moment constant:** $C^* = 64\pi^2/3 \approx 210.55$, identified as a candidate via $16\int_0^\infty t^2/\sinh^2(t/2)\,dt$; the factor 16 is conjectured to arise from the DZT2 ladder normalisation. This is a $\lambda = 5, 7$ benchmark, with $\lambda = 3$ retained as a separate low-bandwidth guardrail and $\lambda = 11$ retained as an open discriminator after the $N = 200$ point. (Part II [4].)

3. **Hecke-normalised gap:** $g_0(m) \approx 2\pi/\log\gamma_m$, classical Montgomery–Odlyzko. (Part III [6].)

2.2 Two Conditional Theorems

- **Theorem B-1** (Part III): Under uniform log-curvature control $H_\lambda(\delta) \le K\delta^2$ with $K \le 1.8 \times 10^{-3}$:

$$A_\lambda(\delta) \le B_\Xi \cdot \exp(K/4) \le 1.0063,$$

  where $B_\Xi = \xi(0)/\xi(\frac{1}{2}) = 1.00579\ldots$ is analytical.

- **Theorem GF1** (Part II): Under eigenmode-weighted kernel convergence producing the effective factor 16:

$$\mathrm{sLI} = \frac{64\pi^2}{3}\,\frac{\lambda}{NL} + o(\cdot).$$

  This is formulated for the non-exceptional converged regime; the $\lambda = 3$ sweep and the open $\lambda = 11$ discriminator prevent an all-λ universal reading.

2.3 Two Isolated Open Proof Slots

1. **B-1 slot:** Uniform log-curvature control from Foundation-Lock structure. Closes Branch B.

2. **GF1 slot:** Kernel convergence + factor-16 identification from DZT2 definitions. Closes Branch A's non-exceptional boundary-moment benchmark, with the low-bandwidth branch kept explicit.

3 Localization Tables

3.1 Li Coefficient Perspective

Step	Statement	Status	Method
S1	Define $\xi(s)$, functional equation	proven	definition
S2	Li coefficients via Bombieri–Lagarias	proven	[1]
S3	Decomposition $\lambda_n = \lambda_n^{(\infty)} + \lambda_n^{(\mathrm{fin})}$	proven	Part I
S4	Archimedean dominance: $\lambda_n^{(\infty)} \sim \frac{n}{2}\log n$	proven	Stirling
S5	$\lambda_n \geq 0$ **for all** $n \geq 1$	**OPEN***	$\Longleftrightarrow$ **RH**
S6	All zeros on $\Re(s) = 1/2$	$\Leftrightarrow$ S5	Li (1997)
S7	Prime counting: $\pi(x) = \mathrm{Li}(x) + O(\sqrt{x}\log x)$	$\Leftarrow$ S6	von Koch

* S5 is equivalent to RH (Li, 1997). The present series gives only conditional reductions.

3.2 Even Dominance Perspective

Step	Statement	Status	Method
E1	Weil explicit formula	proven	Weil (1952)
E2	Weil quadratic form QW_λ	proven	[2]
E3	Even/odd decomposition of QW_λ	proven	Part II
E4	Shift Parity Lemma: each prime favors even	proven	Part II
E5	Gap diagnostics at 33 values, $\lambda \leq 1{,}300{,}000$	evidence / conditional	Part II
E5b	Euler–Maclaurin: $\rho^{\mathrm{EM}} < 0$ for $\lambda \geq 1.2\mathrm{M}$	proven	Part II
E6	**Cumulative step: even dominance for all** λ	**conditional reduction**	$=$ **A6**
E7	Connes' Theorem 6.1 + Hurwitz bridge $\Rightarrow$ RH	conditional on E6	[2]

4 Remaining Open Problems

Remark 4.1 (OP1: Cumulative Even Dominance — reduced to the odd-sector lower bound (v2.2)). The cumulative even dominance problem is reduced to a single spectral question: a quantitative lower bound for $\lambda_1^-(\lambda)$ (Asymptotic Variational Gap Conjecture, Part II).

Remark 4.2 (OP2 resolved: Simplicity of λ_1^+). Certified as simple for all 33 certificate values by interval arithmetic. The intra-even gap grows as $\sim\sqrt{\lambda}$ (Part II).

Conjecture 4.3 (OP3: Analytical Proof of Index Rigidity). *The negative inertia index of K_σ^{sym} is exactly 2 for all N sufficiently large and $\sigma \in (1, \sigma_{\max})$.*

Conjecture 4.4 (OP4: Nuclear Transfer Operator for ξ). *Construct a separable Hilbert space $\mathcal{V}$ and a nuclear family $s \mapsto \mathcal{L}_s$ with $\xi(s)/\xi(0) = \det(I - \mathcal{L}_s)$, spectral radius < 1 for $\Re(s) > 1/2$, and eigenvalue 1 at the zeros.*

Conjecture 4.5 (OP5: Nevanlinna Property of $\log\det_2$). *Is $\log\det_2(I - zR)$ a Nevanlinna function if and only if all zeros lie on the real axis?*

OP2 is resolved. OP1 is reduced to a single asymptotic step. OP3–OP5 remain structural questions.

5 Assessment and Outlook

5.1 What Has Been Achieved

1. A **complete structural analysis** of the gauge-theoretic approach (Part I: R1–R9, K1–K4).
2. The **Shift Parity Lemma** and 33 **computer-assisted finite-range diagnostics** (Part II).
3. The **conditional reduction A1–A8** with analytical backbone (M1$''$, Euler–Maclaurin, PNT transfer) and the GF1 boundary-moment benchmark with $C^* = 64\pi^2/3$ for $\lambda = 5, 7$ (Part II).
4. Two **independent spectral conditional reductions** (Branches B and Z, Part III).
5. The **cross-route synthesis** identifying shared Galerkin pattern and two isolated proof slots (Part IV).
6. **Fifteen excluded paths** (spectral + CCM) and **five methodical dead ends** (Parts III–IV).
7. A 2026 **criticality lens** (RH $\Leftrightarrow \Lambda = 0$, heuristic) re-weighting the routes toward exact-structural mechanisms; a sharper **common-wall** localization (the analytic continuation across $\Re s = 1$); a computable **Laguerre–Pólya/Herglotz diagnostic**; and four closures of *concrete constructions* (not routes) — all without changing the conditional-reduction status (Part IV [3]).

5.2 Current Atlas Status

The series title remains "From Landscape to Atlas: Multi-Route Cartography of an Ongoing Expedition Toward the Riemann Hypothesis." All three living routes are conditional reductions; no single branch has reached conditional-proof level under the project's audit protocol.

Promotion protocol. The title migration from "Atlas" to "From Atlas to Proof" is governed by a five-gate audit: (1) single named hypothesis H; (2) every step from H to RH at theorem-level; (3) two independent reviews including a sceptical refuter; (4) 2–3 day cool-off; (5) explicit Decision Record entry. Default: Atlas remains.

5.3 Future Scenarios

- **One branch dies** (its open hypothesis refuted): documented as dead route; the other two continue.
- **One branch matures unconditionally**: that proof stands as RH; other branches retained for independent verification.
- **Both/all three mature**: rare, but mathematically valuable as multi-route verification.
- **None matures, but partial progress continues**: research programme continues; the Landscape preserves the audit. This is the default outlook.

5.4 Outlook

Concrete next steps (not time-bound):

- Rigorous derivation of the two open proof slots (B-1 log-curvature, GF1 factor-16).
- Resolution of the $\lambda = 11$ GF1 discriminator, validation at higher λ, and a theorem-level separation of the $\lambda = 3$ low-bandwidth branch.
- κ_∞-asymptotics for $\lambda \to \infty$.
- Promotion-gate audit if a slot closes.
- Develop the exact-structural lines (M = Bost–Connes $\leftrightarrow$ Meyer; P-A = global adelic trace route) that the 2026 criticality lens elevates, in parallel with the two margin-type proof slots (Part IV [3]).

6 Connections to the Wider Landscape

6.1 Connes' Program

The work in this series fits directly into Connes' spectral approach to RH [2]. The Shift Parity Lemma (Part II) provides analytical evidence for the even dominance hypothesis (Theorem 6.1). The I-1 Normal-Family reframe (Part III) decomposes Wall (ii) into a single named condition (ii-a). The CCM Microcluster Programme (Part III) attacks the same wall through the Fourier model.

6.2 Nyman–Beurling–Baez-Duarte

The Li criterion, the Weil positivity criterion, the Nyman–Beurling criterion, and the Baez-Duarte criterion are all equivalent to RH. Direct implications between them (without passing through RH) are known only in two cases: Weil $\Rightarrow$ Li ([1]) and Nyman–Beurling $\Rightarrow$ Baez-Duarte (trivial).

6.3 The Trace-Class Obstruction as Universal Pattern

Obstruction K4 (trace-class failure at $\Re(s) = 1/2$, Part I) appears across QFT (UV divergences), statistical mechanics (thermodynamic limit), and NCG (zeta-function regularisation). The Even Dominance approach circumvents this by working with finite-dimensional Galerkin blocks rather than infinite-dimensional transfer operators.

7 Full Summary Table

Contribution	Status	Paper
Thermodynamic formalism for ζ	established	I
9 unconditional structural results (R1–R9)	proven	I
4 excluded paths (K1–K4)	proven	I
Shift Parity Lemma	proven (analytic)	II
Gap diagnostics (33 values, $\lambda \leq 1{,}300{,}000$)	evidence / conditional	II
Leading-Mode Cancellation ($c = 2 + \sqrt{2}$)	proven	II
Euler–Maclaurin: $\rho^{\mathrm{EM}}(13) < 0$	proven	II
GF1 benchmark: sLI $\sim C^{*}\lambda/(NL)$ for $\lambda = 5, 7$; $\lambda = 3$ separate, $\lambda = 11$ open	conditional (Thm GF1)	II
I-1 Normal-Family Reframe: (ii) $\to$ (ii-a)	rigorous	III
Foundation-Lock + converged empirics	rigorous + numerical	III
Theorem B-1: $A_\lambda \leq 1.0063$ (conditional)	conditional	III
c_Ξ Hadamard curvature (RH-neutral)	analytical	III
CCM three C2ca-inputs	conditional reduction	III
6 excluded spectral + 9 excluded CCM paths	documented	III
Cross-route Galerkin pattern	empirical	IV
11 systematically explored alternatives	documented	IV
Hypothesis correction chain (GF1 Iter 15–27)	documented	IV
Cumulative step (A6)	**conditional reduction (v2.2)**	II
Riemann Hypothesis	**conditional reduction (v2.2)**	I–V

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AI Disclosure

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